ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2024–2025**

DURATION: 2 HOURS **FULL MARKS: 120**

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Jordan, a passionate sneakerhead, wants to crack the code behind the pricing of trendy new sneakers. He suspects that the price of a sneaker depends on two main factors: (a) the comfort rating, which measures how comfortable the shoe feels during wear, and (b) the hype score, which reflects how popular and sought-after the sneaker is in the sneakerhead community. To test his theory, Jordan collected data on four of the latest sneaker releases shown in Table 1.

Table 1: A dataset of sneaker prices based on comfort rating and hype score.

Sneaker	Comfort Rating (x_1)	Hype Score (x_2)	Price (USD) (y)
Nike Air Force 1	8	7	180
Adidas Superstar	6	9	210
Converse	7	6	165
Vans Old Skool	9	8	220

Answer the following questions based on the dataset in Table 1 assuming that the price y depends linearly on the comfort rating and hype score, i.e.,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

a) Formulate the problem of finding the best-fitting line in $Ax = b$ form where A, b, and x are the coefficient matrix, constant vector, and variable vector respectively, and determine the following: 12 (CO3) (PO1)

   i. The coefficients of the best fit line (β_0, β_1, and β_2) as $\hat{x}$.
   ii. The projection of all data (p) onto the column space of A.
   iii. The projection matrix P that projects b onto the column space of A.
   iv. The error vector e and the mean squared error (MSE) of the fit.

b) An outlier is a data point that significantly deviates from the overall pattern of the data. Curious to verify the results obtained in Question 1.a, Jordan decided to expand his dataset by collecting information on a few more sneakers. While reviewing the data of the new sneakers, he noticed that the price of one sneaker seemed unusually different from what his regression model would predict, which he suspects to be an outlier. The new dataset is shown in Table 2. Identify which data point is most likely the outlier and justify your answer. Your conclusion must be supported by appropriate calculations based on results obtained in Question 1.a. 8 (CO3) (PO1)

Table 2: A dataset of prices of new sneakers.

Sneaker	Comfort Rating (x_1)	Hype Score (x_2)	Price (USD) (y)
New Balance 990	9	6	175
Nike Air Jordan 1	8	12	295
Abidas UltraBoost	5	9	360
Nike Dunk	7	9	225

c) Do you think the regression model should depend on both the comfort rating and the hype score, or would a single factor be sufficient? If a single factor is sufficient, determine which one provides the better model. Justify your answer mathematically by comparing the fitting errors using the data given in Table 1. 10 (CO3) (PO1)

Assume that a regression model using a single factor has the form

$$y = \beta_0 + \beta_1 x_i + \beta_2 x_i^2,$$

where x_i represents either the comfort rating or the hype score. Compute the corresponding errors and use them to support your conclusion. Do not include the data from Table 2 in your calculations.

2. a) Find Singular Value Decomposition (SVD) of the following matrix A and express it in $A = U\Sigma V^T$ form, identifying the matrices U, Σ and V. 15 (CO3) (PO1)

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

b) A sequence is given by 2 + 7 + 1 (CO3) (PO1)

$$-1,\ 3,\ 5,\ 27,\ 101,\ 411,\ 1637,\ \cdots$$

where each term is obtained as a linear combination of the two preceding terms. The first two terms of the sequence are $a_0 = -1$ and $a_1 = 3$.

i. Determine the recurrence relation for the sequence. In other words, express a_n in terms of a_{n-1} and a_{n-2}.

ii. Express the recurrence relation obtained in Question 2.b)i in Matrix form and use diagonalization to derive a formula for the k-th term of the sequence.

iii. Using the result obtained in Question 2.b)ii, compute the 10th term of the sequence.

c) Given, the bases of Columnspace, $C(D)$ and Left Nullspace, $N(D^T)$ of matrix D in $\mathbb{R}^5$. Determine the values of p,q and r. 5 (CO3) (PO1)

Table 3: Bases of $C(A)$ and $N(A^T)$

Columnspace, $C(D)$	Left Nullspace, $N(D^T)$
$\begin{bmatrix} 1 \\ 2 \\ p \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} q \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ -4 \\ -3 \\ 0 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 4 \\ 0 \\ r \\ 1 \end{bmatrix}$

3. a) Convert the given vectors ($\vec{v_1}, \vec{v_2}, \vec{v_3}$ and $\vec{v_4}$) into orthonormal vectors using the Gram- Schmidt process. Verify your result by showing that the dot product of any two vectors in the obtained set is zero. 12 (CO3) (PO1)

$$\vec{v_1} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix}, \vec{v_2} = \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \vec{v_3} = \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix}, \vec{v_4} = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

b) Show that if λ is an eigenvalue of an invertible matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} and $\mathbf{x}$ is a corresponding eigenvector (without using a mathematical example). 6 (CO3) (PO1)

c) In a 3-dimensional vector space, determine whether an orthogonal complement subspace exists in each of the following cases. If it does not exist, briefly explain why. If it does exist, describe the nature of the orthogonal complement. 3 + 3 (CO3) (PO1)

i. A subspace spanned by three mutually orthogonal vectors in the vector space.

ii. A plane spanned by two linearly independent vectors.

d) The eigenvalues and eigenvectors of a matrix M are $\lambda_1 = 2$ and $\lambda_2 = 4$ with corresponding eigenvectors $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively. What should be the nature of the matrix M based on the given information? Determine matrix M, and verify whether your assumption is correct. 6 (CO3) (PO1)

4. a) Given, $Ax = b$, for 10 (CO2) (PO1)

$$A = \begin{bmatrix} -2 & 1 & 5 \\ 3 & 2 & 7 \\ -5 & 3 & 9 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

where,

$$det(A_1) = 54, det(A_2) = -189, det(A_3) = 27$$

A_i is the matrix similar to matrix A but its i-th column is replaced by vector b, for example,

$$A_i = \begin{bmatrix} b_1 & 1 & 5 \\ b_2 & 2 & 7 \\ b_3 & 3 & 9 \end{bmatrix}$$

Retrieve vector x and b using your knowledge of Matrices, Determinants and Cramer's rule.

b) Given, 10 (CO2) (PO1)

$$B = \begin{bmatrix} 3 & -2 & 0 & -4 \\ 5 & -6 & 7 & -8 \\ -1 & 2 & -3 & 4 \\ -5 & 6 & -7 & 8 \end{bmatrix}$$

Apply the False Expansion Theorem on matrix B to show that

$$\sum_{i=1}^{n} b_{xi} C_{yi} = 0 \quad (\text{where } x \neq y)$$

While solving, clearly mention each b_{xi} and C_{yi} that you use. Here, x and y represent row numbers, n represents the total number of rows/columns, b_{ij} represents the entry in the i-th row and j-th column of matrix B, and C_{ij} represents the cofactor of entry b_{ij}.)

c) Determine the determinant of matrix C using the Big Formula. 10 (CO2) (PO1)

$$C = \begin{bmatrix} -2 & 0 & 3 & 0 \\ 0 & 7 & 1 & 0 \\ 4 & 0 & 0 & -5 \\ 0 & -9 & 0 & 6 \end{bmatrix}$$

Appendix for Math 4341

A Orthogonality

$C(A) \perp N(A^T)$ in $\mathbf{R^n}$ (for $A_{n\times m}$)

$C(A^T) \perp N(A)$ in $\mathbf{R^m}$ (for $A_{n\times m}$)

If $\vec{a} \perp \vec{b}$, then $|\vec{a} \cdot \vec{b}| = 0$

B Projection and LSR

$\hat{x} = (A^T A)^{-1} A^T \vec{b}$

$p = A\hat{x}$ (projection of $\vec{b}$ onto $C(A)$)

$P = A(A^T A)^{-1} A^T$ (projection matrix)

$e = \vec{b} - p$ (error vector)

$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} e_i^2$

$P = P^2 = P^T$ (Orthogonal Projection Matrix)

C Gram Schmidt Process

$q_i^T q_j = 0$ for $i \neq j$

$q_i^T q_i = 1$

$Q^T Q = I$ or $Q^{-1} = Q^T$

If $b \in C(A)$, then projection $p = b$

$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j$, $k \geq 2$

$e_k = \frac{u_k}{|u_k|}$, $k \geq 1$

e.g. $C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$,

$C_{norm} = \frac{C}{|C|}$

D Determinants

$\sum_{j=1}^{n} a_{kj} C_{ij} = \begin{cases} \det(A), & k = i \\ 0 & k \neq i \end{cases}$

$A^{-1} = \frac{1}{\det(A)} \text{CoF}(A)^T$

$x_i = \frac{\det(A_i)}{\det(A)}$

E Eigenvalues & Eigenvectors

$Ax = \lambda x$ (A square)

$\det(A - \lambda I) = 0$

F Diagonalization

$A = S\Lambda S^{-1}$

$A^k = S\Lambda^k S^{-1}$

$A = PDP^{-1}$ (D diagonal eigenvalues, P eigenvectors)

G Symmetric Matrices & SVD

$A = Q\Lambda Q^T$

$A = \sum_{i=1}^{n} \lambda_i q_i q_i^T$

$A = U\Sigma V^T$

$A^T A = V(\Sigma^T \Sigma)V^T = V\Lambda V^T$

$AA^T = U(\Sigma\Sigma^T)U^T = U\Lambda U^T$

$Av_i = \sigma_i u_i$